DAY 1

SKILLBUILDER 2

Tokens and Embeddings in NLP

Task 1: Concept Explanation

1. What is a Token?

A token is the smallest unit of text processed by an NLP model. A token can be a word, subword, punctuation mark, or character depending on the tokenizer.

2. Tokenization Example

Sentence: "AI is transforming industries"
Tokens:
["AI", "is", "transforming", "industries"]

Task 2: Embedding Understanding

1. What are Embeddings?

Embeddings are numerical vector representations of words, sentences, or documents. They capture semantic meaning so that text with similar meanings has similar vector representations.

2. Why are Embeddings Important for LLMs?

Embeddings allow language models to understand relationships between words, capture context, compare meanings, retrieve relevant information, and generate more accurate responses.

Task 3: Application

1. How Embeddings Help

Similarity Search:
Embeddings make it possible to find documents or questions with similar meanings even when they use different words.

Text Understanding:
Embeddings help models recognize context, intent, and relationships between

words, improving classification, summarization, translation, and question answering.

2. Real-World Example

A customer support chatbot converts user questions and knowledge-base articles into embeddings. It retrieves the most relevant article based on semantic similarity and generates an accurate answer.